



Dr. Nadeem Tariq Beigh

Nationality: Indian

Age: 34 Years

Marital status: Married

Location: Abu Dhabi

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RESEARCH INTERESTS

Sustainable Sensing, Printed Sensors, Ink optimizations, Self-Powered Systems, Energy Harvesters, MEMS/NEMS, Micro/Nano Fabrication, and Simulation

Dr. Nadeem Tariq Beigh is an interdisciplinary researcher working at the intersection of flexible and wearable electronics, printed sensors, multifunctional nanocomposites, MEMS/NEMS, self-powered systems, and AI-enabled sensing. He is currently a **Visiting Researcher at the Indian Institute of Technology Delhi**, where he continues to advance research in intelligent microsystems, energy harvesting, and next-generation sensing technologies.

He recently completed his **postdoctoral research at the Smart and Intelligent Sensors Laboratory, Khalifa University, Abu Dhabi**, where he developed low-cost flexible wearable platforms, porous nanocomposite pressure sensors, conductive ink-based printed devices, and multifunctional materials for multimodal sensing. His work also explored nonlinear PMUT dynamics and reservoir computing, leading to several TRL-4 prototypes, patent applications, and commercialization-oriented technologies for healthcare and AgriTech.

Previously, he worked at **Nanyang Technological University, Singapore**, on MEMS-enabled terahertz switching for high-frequency communication systems. He received his **Ph.D. in Electrical Engineering from IIT Delhi**, where he developed photopatternable piezoelectric-triboelectric nanocomposites and self-powered IoT sensing platforms. His research vision is to translate advanced materials and microsystems into scalable, intelligent, and sustainable technologies for healthcare, precision agriculture, robotics, and Industry 5.0.

EDUCATION

[01/19 - 03/24] Doctor of Philosophy in Electrical Engineering
Department of Electrical Engineering, Indian Institute of Technology, New Delhi, India
Advisor: [Prof. DHIMAN MALLICK](#)
Thesis Title: **Development of Dual Piezoelectric/Triboelectric Nanocomposites for Sensing And Energy Harvesting Applications**
CGPA: 9.12/10

[06/16 - 08/18] Masters of Technology in VLSI DESIGN
School of Engineering Technology, Sharda University, Delhi
Advisor: Prof. SABYASACHI MUKHOPADHYAY
Thesis Title: **Design of High Throughput Sigma-Delta ADCs for Highly Sensitive Signal Digitization.**
CGPA: 9.45/10

[06/11 - 08/15] Bachelor of Technology in ELECTRONICS COMMUNICATION ENGINEERING
Department of Electronics and Communication Engineering, Islamic University of Science Technology, JK

Advisor: DR. FAROZE AHMAD
Thesis Title: **Design of Digitally controlled Filters.**
CGPA: 7.85/10

WORK EXPERIENCE

07/26 - CURRENT	VISITING RESEARCHER IN IML Department of Electrical Engineering, IIT Delhi, New Delhi Advisor: Line Manager: PROF.DHIMAN MALLICK
08/24 - 07/26	POST DOCTORAL RESEARCHER IN SIS-LABS Department of Mechanical and Nuclear Engineering, Khalifa University, Abu Dhabi Line Manager: Dr. NOUHA ALCHEIKH
02/24 - 08/24	RESEARCH FELLOW IN I-LABS School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore. Line Manager: Dr. MOHAMMAD SAMIZADEH NIKOO
01/19- 01/24	RESEARCH ASSISTANT Nano research facility (NRF) IIT Delhi, New Delhi, India
12/17- 05/18	RESEARCH INTERN Department of Electronics & Communication NIT Srinagar, J&k, India

RESEARCH PROJECTS

2019-22	DEVELOPMENT OF MICRO-ENERGY HARVESTING DEVICES- IRD IIT Delhi - Studied the % weight ratio based optimization of SU-8 nanocomposites and analysed their structural, morphological, optical and electrical properties. - Validated the usability of these novel nanocomposites by extensively investigating the XRD, SEM, PFM and UV-VIS to nominate them as alternate piezoelectric and triboelectric thin films over the conventional ceramic based thin films. - The weight ratio based optimization of these nanocomposite provided an optimum value for added nano-particles to maintain good photo-patternability, piezoelectric and triboelectric response response. - Development of rapid design cantilever and bending prototypes for biomechanical energy harvesting. Published in IEEE Sensor Journal , Advanced Electronic. Materials
2020-23	HIGH EFFICIENCY MICRO/NANO ENERGY HARVESTING SYSTEM FOR SELF-POWERED SMART ENVIRONMENT- DST-RCN (India-Norway Joint Project) - The development of novel solutions to impact induced loss in piezoelectric impact driven energy harvesters. - Development of alternate energy routes via dual transduction nanocomposites to improve the efficiency by compensating the impact loss by inherent triboelectric nature of polymeric nanocomposites. - Proposed an alternate design strategies in impact driven piezoelectric energy harvesters for a high performance, wide band operation. Published in Physical Review Energy

2021-23	SELF-POWERED SENSOR SYSTEM FOR SOFT ROBOTICS AND HUMAN-MACHINE INTERACTION (HMI)-I-Hub Foundation for Cobotics, Technology Innovation Hub, IIT Delhi • Development of a smart sensors by employing multi-modal transduction mechanism to sense multiple sensory responses via a single device. • The complete bottom up design approach incorporating sensor design on purely flexible and conformal to the body worn attachments. • Integration of machine learning algorithms to improve the event classification for an effective smart sensor operation. Published in IEEE MEMS2023, Nano Energy
2022-23	DEVELOPMENT OF HIGH G ELECTROMAGNETIC ENERGY HARVESTER FOR RAILWAY APPLICATION.IITD-FITT • Development a high impact, high g electromagnetic energy harvester powered from the mechanical vibrations from the train motion to drive low power sensor system for real-time condition monitoring applications. • Bottom up design, simulation and experimental validation of tubular structured high g energy harvesters wit shock resistant housing.
2020-22	HARDWARE REALIZATION FOR NEUROMORPHIC COMPUTING USING MEMS OSCILLATOR NETWORKS-MFIRP IIT Delhi with NCTU Taiwan • Development of nonlinear resonators based on PiezoMUMPs and SOIMUMPS fabrication process flow for energy harvesting/sensing applications. • Design of auto-parametric inter-mode energy injecting resonators capable of reservoir computation and neuromorphic applications.
2020-22	DEVELOPMENT OF OPTIMIZED PIEZOELECTRIC THIN FILMS CMOS/BIO COMPATIBLE ENERGY HARVESTING AND SENSING APPLICATIONS. • Exploring the optimization of RF sputtered AlN and spin coated PVDF thin films and their respective nanocomposites on a variety of substrates for flexible, CMOS/Bio compatible energy harvesting and sensing applications. • Developing the high performance nanocomposites with piezoelectric base matrix with a high β phase and c-axis orientation.
2024	ULTRA HIGH SPEED ELECTRONICS FOR THZ APPLICATIONS • Unifying M/NEMS and RF electronics for ultra high speed, near zero power THz switching applications and reconfigurable M/NEMS metasurfaces for THz manipulations.
2024-	SINGLE MICRO SENSOR FOR MULTI-PARAMETER SENSING • Exploring mode coupling and nonlinear behavior of N/MEMS resonators for multi-parameter sensing for a low-power, smart sensor systems.
2024-	SMART AND INTEGRATED SELF-POWERED WEARABLES FOR BIOMECHANICAL SENSING • Development of fully flexible, self-powered biomechanical sensors with integrated, flexible readout circuits. • Exploring polymer-filler combinations for a bio-compatible and large area sensing applications. • Design of body-conformal and compatible circuit interface with integrated power management, ADCs and MCUs.

2025-	FLEXIBLE IoT SYSTEMS FOR PLANT MICROCLIMATE • Development of 2D material composite based fully flexible and integrated, flexible readout circuits for temperature, humidity and gas sensors for the micro-climate in a plant environment.
2025-	DEVELOPMENT OF HIGH PRECISION TEMPERATURE SENSOR USING NANOMECHANICAL DEVICES FOR MID-IR DETECTOR. IITD/IITD-AD/KU

STUDENT MENTORING

2019-23	MENTORING STUDENTS FOR MASTER THESIS AND BACHELOR'S PROJECTS • Project 1: Nonlinear MEMS beam resonator design. • Project 2: Magnetoelectrics for wireless power transfer in bio-implantable devices. • Project 3: Flexible, wireless, gait sensors. • Project 4: Neuromorphic MEMS. • Project 5: Genetic algorithm driven design of MEMS energy harvesters.
2024-26	MENTORING STUDENTS FOR MASTER RESEARCH COURSE AND UNDERGRADUATE SDPs • Project 1: Exploring self-powered sensor systems for Plant micro-climate study (KU-SDP). • Project 2: A smart and fully integrated flexible insole for critical patient monitoring (KU-SDP). • Project 3: Smart Colorimetric sensing of food packages (KU-Mubadala). • Project 4: Smart Agricultural Drone For Early Detection Of Red Palm Weevil (KU-SDP).

COMMERCIALIZATION

- **NanoMap:** A smart insole for gait analysis based on US-patent- Reaches TRL-4 and is progressing to TRL-7 with KUEC funding-2026.
- **HNK1:** A smart humidity sensors system- Reaches TRL-4 and is progressing to TRL-7 with KUEC funding-2026.
- **ESenSKin:** An multi-sensing e-SKin for robotics applications-Reaches TRL-4 and is progressing to TRL-7 with KUEC funding-2026.

PUBLICATIONS

PATENTS

- A.Bashir, J.D.Dixit, I.Murtaza, F.A.Bhat, **N. T. Beigh**, N.Ahmad, *Self-Propelled Boom Sprayer and Methods Thereof*. Indian Patent-Application No. 202011042265.
- **N. T. Beigh**, N. Alcheikh *Rapid Fabrication of Porous Polymer Nanocomposites for Multi-Sensing Applications*-US Patent-Application No. 19244851.
- **N. T. Beigh**, F. Shahzad, N. Alcheikh *Formulation of a PDMS/MXene Ink for Electrohydrodynamic Printing of Stable Temperature Sensors*-U.S. Provisional Application No. 63/053,754).
- **N. T. Beigh**, M. Al Hadi, N. Alcheikh *Printable Elastomeric Nanocomposite for Multifunctional Pressure, Strain, and Temperature Sensing*-US Patent-Under Review (Invention disclosure submitted).
- **N. T. Beigh**, N. Alcheikh *DMSO-Activated TiCT MXene Ink Compositions, Methods of Making the Same, and Printed Conductive Articles Produced Therefrom*-US Patent-Under Review (Invention

disclosure submitted).

- **N. T. Beigh**, F.T.Beigh, D.Mallick *Machine Learning Enabled Hind Foot Deformity Detection using Individually Addressable Hybrid Pressure Sensor Matrix*-Indian Patent-Under review (Invention disclosure submitted).

Book Chapter

- D.Mallick, S.Naval, **N. T. Beigh**, MEMS-based energy harvesting devices: Overview of recent progress,in *MEMS Applications in Electronics and Engineering*, (AIP Publishing, New York, 2022) [Link](#).
- S.Naval, **N. T. Beigh**, A. Jain, D. Mallick SU-8/BTO Nanocomposite Cantilever Beam based Triboelectric Generator.

Selected Publications

- [J1] **N. T. Beigh**, Nouha Alcheikh. Vapor-Induced Porosity in Graphene/PDMS: A Scalable Route to High-Performance Pressure Sensors *Nature Microsystems & Nanoengineering*, **11(181)** (2025). (Impact factor: 9.9; Citations: 1) [Link](#)
- [J2] **N. T. Beigh**, S.Singh, A.Goswami, D.Mallick. Dual Piezoelectric/Triboelectric Behavior of BTO/SU-8 Photopatternable Nanocomposites for Highly Efficient Mechanical Energy Harvesting. *Advanced Electronic Materials*, **8(10)**, 2200338 (2022).(Impact factor: 5.3; Citations: 16) [Link](#)
- [J3] **N. T. Beigh**, S.Naval, D.Mallick. Impact-loss-compensated wideband vibrational energy harvesting using a hybrid BaTiO₃/SU-8 nanocomposite as the active layer. *Physical Review Energy*, **1(3)**, 033004 (2022).(Impact factor: ; Citations: 8) [Link](#)
- [J4] S.Naval, **N. T. Beigh**, D.Mukherjee, D.Mallick. Multi-output AC/DC Triboelectric Generator with Dual Rectification, *Nano Energy*, **105**, 108004 (2022).(Impact factor: 17.1 ; Citations: 19) [Link](#)
- [C1] **N. T. Beigh**, F.T.Beigh, D. Mukherjee, S.Naval, D. Mallick , Machine Learning Enabled Hind Foot Deformity Detection Using Individually Addressable Hybrid Pressure Sensor Matrix. In *2023 IEEE 36th International Conference on Micro Electro Mechanical Systems (MEMS)*, (pp. **343-346**). IEEE (2023). [Link](#)
- [J5] **N. T. Beigh**, F.T.Beigh, D.Mallick. Machine Learning Assisted Hybrid Transduction Nanocomposite Based Flexible Pressure Sensor Matrix for Human Gait Analysis *Nano Energy*, **116**, 108824 (2023). (Impact factor: 17.1 ; Citations: 59) [Link](#)
- [J6] **N. T. Beigh**, S.S.Chauhan, D.Mallick. High -Phase Barium Titanium Oxide Nanoparticle/P(VDF-TrFE) Composites for Mechanical Sensing *ACS Applied Nanomaterials*, **7(24)**,28106–28118 (2024).(Impact factor: 5.5 ; Citations: 5) [Link](#)

Journal papers

Under Review

- [R1] **N. T. Beigh**, F. Shahzad, N. Alcheikh. Electrohydrodynamic Printing of PDMS/ Ti₃C₂T_x MXene Hybrid Inks Toward Printable and Flexible Sensors *Under Review- Chemical Engineering Journal*
- [R2] **N. T. Beigh**, M.Al Hadi, N. Alcheikh. A Flexible Multimodal MWCNT/PEDOT: PSS/Ecoflex Sensor for Decoupled Temperature and Pressure Monitoring via AI-Assisted Signal Processing *Under Review- Nature Microsystems and Nanoengineering*

- [R3] H. Nawaz, **N. T. Beigh**, R.Pandey, S.W.Hasan, N. Alcheikh. A Flexible MOF–MXene Based Wearable Sensing Platform for Wireless Temperature–Humidity Monitoring of Human Respiratory Conditions *Under Review- Advanced Powder Materials*
- [R4] **N. T. Beigh**, R.Pandey, M. Hemid, H. Nawaz, C.Abuoudah, L.Zheng, S.W.Hasan, N. Alcheikh. MXene/MOF Hybrid Materials for Sensing: Interfacial Mechanisms, Transduction Pathways, and Design Principles *Under Review- Advanced Composites and Hybrid Materials*
- [R5] F. T. Beigh, **N. T. Beigh**, Dhiman Mallick. Design-Guided Laser-Induced Graphene for Multifunctional Thermal Platforms: From Heating to Sensing and Wettability Control *Under Review in Nanoscale* (2026).

Published

- [J7] F. Arshad, **N. T. Beigh**, G. A. Naikoo, N. Alcheikh. Beyond Glucose: Intelligent, Multi-Analyte, and AI-integrated Biosensing Systems for Next-Generation Diabetes Management *Sensor and Actuators Reports*, **12**, 100481, (2026). [Link](#)
- [J8] F. T. Beigh, **N. T. Beigh**, Dhiman Mallick. Parametrically Engineered Laser-Induced Graphene with Tunable Electrical and Morphological Properties for Flexible Sensor Applications *Accepted in ACS Omega* (2026).(Impact factor: 5.2; Citations:) [Link](#)
- [J9] H.Amara, **N. T. Beigh**, N. Alcheikh. Dynamic Analysis of a Thermally Multi-Input Guided Microbeam for High-Performance, Wide-Range Magnetic Field Sensing *IEEE Sensors Journal*, **26(3)**, 3986-3995, (2026).(Impact factor: 5.1; Citations:) [Link](#)
- [J10] H.Amara, **N. T. Beigh**, N. Alcheikh. Smart Resonant Micro-sensor and Micro-actuator: High-Performance, Wide Range Bi-Axial Magnetic Sensitive/ Insensitive Micro-Device for Multifunctional Sensing Applications *Nature Microsystems & Nanoengineering*, **11(174)** (2025).(Impact factor: 9.9; Citations: 1) [Link](#)
- [J11] S. Singh, **N. T. Beigh**, P. Gupta, A. Goswami, D.Mallick. Polarity Dependent Surface Charge Retention on CYTOP Fluoropolymer for Durable TENG Applications *Materials Today*, **43**,102656 (2025).(Impact factor: 6.9; Citations: 1) [Link](#)
- [J12] D. Mukherjee, S. Singh, **N. T. Beigh**, S. Naval, P. Pathak, A. Goswami, D.Mallick. A Metglas Based Magneto-Triboelectric Energy Harvesting Device for Biomedical Applications *IEEE Sensors Journal*, **24(19)**, 29874-29883 (2024). (Impact factor: 5.1; Citations: 4) [Link](#)
- [J13] **N. T. Beigh**, N. Murali, F.T. Beigh, S. Betal, D.Mallick. Magnetically Polarizable Energy-Efficient Magnetoelectric Core-Shell Nanocomposites For Magnetic Field Sensing *IEEE Sensors Journal*, **24(15)**, 23613-23621 (2024). (Impact factor: 5.1; Citations: 3) [Link](#)
- [J14] **N. T. Beigh**, D.Mallick. Low-Cost, High-Performance Piezoelectric Nanocomposite for Mechanical Energy Harvesting. *IEEE Sensors Journal*, **21(19)**, 21268-21276 (2021).(Impact factor: 5.1; Citations: 24) [Link](#)
- [J15] S.Naval, **N. T. Beigh**, A Jain, D.Mallick. Flexible V-Shaped Piezoelectric-Triboelectric Device for Biomechanical Energy Harvesting and Sensing. *Journal of Physics D: Applied Physics*, **55(36)**,365501 (2022).(Impact factor: 3.2; Citations: 26) [Link](#)
- [J16] S.Naval, **N. T. Beigh**, A.Jain, D.Mallick. Bandwidth Tunable Vibration Energy Harvester based on Hybrid Triboelectric-Piezoelectric Array. *Engineering Research Express*, **4(4)**,045022 (2022). (Im-

pact factor: 1.6; Citations: 16) [Link](#)

- [J17] M.Yousuf, **N. T. Beigh**, D.S.Arya, M.Garg, Pushpapraj Singh, D. Mallick. A Sensitive and Flexible Poroelastic Barium Titanate Matrix for Pressure Sensing Applications. *IEEE Sensor Letters*, **7(2)**, 1-4.(2023).(Impact factor: 2.2; Citations: 8) [Link](#)
- [J18] F.T.Beigh, S. Naval, V.Garg, **N. T. Beigh**,D. Mallick. Intelligent Workplace Activity Monitoring and Detection using Self-powered Triboelectric/Piezoelectric Sensor Augmented Machine Learning.*IEEE Sensor Letters*, **7(6)**, 1-4.(2023). (Impact factor: 2.2; Citations: 13) [Link](#)

Conferences Papers

- [C1] **N.T.Beigh**,M. A. Hadi, N. Alcheikh. Fully Printed MWCNT/PEDOT:PSS/Ecoflex Tactile–Thermal Sensor with Dual-Mode Piezoresistive and Thermoresistive Transduction *Poster Presentation at EuroSensors2026, Zurich..*
- [C2] H. Nawaz, **N.T.Beigh**, N. Alcheikh. Flexible Wearable MXene-Based Printed Temperature Sensor for Multifunctional Applications *Poster Presentation at EuroSensors2026, Zurich..*
- [C3] **N.T.Beigh**, N. Alcheikh. Vapor-Channeled Porous Pressure Sensors *Oral Presentation at EuroSensors2025, Poland.*[Link](#).
- [C4] H. Amara, **N.T.Beigh**, N. Alcheikh. High-Performance Mems Magnetic Sensor Based on a Smart Tunable Resonator *Poster Presentation at IEEE MEMS2025.*[Link](#).
- [C5] F.T.Beigh, **N.T.Beigh**, D.Mallick.High-Performance Flexible Heaters with Optimum Laser Induced Graphene Parameters *Oral Presentation at IEEE FELPS2024.*[Link](#).
- [C6] **N.T.Beigh**, F.T.Beigh, D.Mallick.Leveraging Photopatternable Nanocomposites for High Performance Tactile Neuromorphic Sensing *Oral Presentation at IEEE EDTM2024.*[Link](#).
- [C7] F.T.Beigh, Y.C.Chuang, P.Singh,**N.T.Beigh**,S. Narain, S.Singla, Yi Chiu, D.Mallick.Reservoir Computing with a MEMS Nonlinear Resonator for In-Sensor Computing *Oral Presentation at IEEE EDTM2024.*[Link](#).
- [C8] F.T.Beigh, **N. T. Beigh**, D.Mallick, Optimization Strategies for Laser-Induced Graphene on Various Substrates for Flexible Strain Sensing . *In IEEE APSCON2024* (2023) (Accepted-Lecture Presentation). [Link](#).
- [C9] D. Mukherjee, S,Naval, **N. T. Beigh**, D.Mallick, Metglas Based Multi-Modal Sensing Employing Magnetostrictive and Triboelectric Properties. *In IEEE Sensors2023* (2023) [Link](#).
- [C10] F.T.Beigh, **N. T. Beigh**, D.Mallick, Localized Biomechanical Strain Sensing with Graphene/SU-8 Nanocomposite Based Highly Flexible Piezoressistive Sensor. *In IEEE Transducers2023* (2023) (Accepted-Oral Presentation).
- [C11] S.S.Chauhan, **N. T. Beigh**, D.Mukherjee, D.Mallick, Development and Optimization of Highly Piezoelectric BTO/PVDF-TrFE Nanocomposite Film for Energy Harvesting Application. *In 2022 6th IEEE International Conference on Emerging Electronics (ICEE)* (2022).[Link](#)
- [C12] S.S.Chauhan, **N. T. Beigh**, D.Mukherjee, D Mallick, Flexible Vibrational Energy Harvester with Groove Design Using BTO/PVDF-TrFE Film for Higher Output. *In the IEEE APSCON 2023*

(2023).

[Link](#)

- [C13] S. Naval, P. Verma, **N. T. Beigh**, A.Jain, D.Mallick, Flexible DC Triboelectric Generator with Associated Conditioning Circuit. *In 2022 21st International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS)*, (pp. **30-33**). IEEE (2022). [Link](#)
- [C14] S. Naval, **N. T. Beigh**, A.Jain, D.Mallick, Comparative study of frequency response of triboelectric and piezoelectric energy harvesters. *In 2021 IEEE 20th International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS)*, (pp. **224-227**). IEEE (2022). [Link](#)
- [C15] **N. T. Beigh**, D.Mallick. Highly Efficient Low Threshold Parametrically Forced MEMS Vibrational Energy Harvesters. *In 2020 5th IEEE International Conference on Emerging Electronics (ICEE)*, (pp. **1-4**). IEEE (2020). [Link](#)
- [C16] **N. T. Beigh** , D.Mallick, Bistable Piezoelectric Energy Harvesters for Low Frequency, Low G Applications. *In 2020 5th IEEE International Conference on Emerging Electronics (ICEE)*. IEEE (2020).
- [C17] **N. T. Beigh**, D. Mallick, Optimization of ZnO/Su-8 Based Photopatternable, Piezoelectric Nano-Composites for Mechanical Energy Harvesting Applications. *In 2021 21st International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers)*, (pp. **875-879**). IEEE (2021). [Link](#)
- [C18] **N. T. Beigh**, D. Mallick, Design of Nonlinear MEMS Buckled Beam Structures for Low-threshold Mechanical Energy Harvesting. *In International Conference on Precision, Meso, Micro and Nano Engineering (COPEN11)* (2019). [Link](#)
- [C19] **N. T. Beigh**, P S Azad, P Parkash, D. Mallick, High Performance, Nonlinear Piezoelectric MEMS Energy Harvesting from Low-threshold Mechanical Vibrations. *In 2019 19th International Conference on Micro and Nanotechnology for Power Generation and Energy Conversion Applications (PowerMEMS)*,(pp. **1-5**). IEEE(2020) [Link](#)
- [C20] Shalini, D.S. Arya, **N. T. Beigh**, S. Kumar, R. Mudgal, P. Singh, P.K. Muduli, D. Mallick, Ankur Goswami, Energy Harvesting From Water Droplet Motion Confined On A Hydrophobic-Hydrophilic Stripped Surface. *In 2022 IEEE 35th International Conference on Micro Electro Mechanical Systems Conference (MEMS)*,(pp. **943-946**). IEEE (2022). [Link](#)
- [C21] F.T. Beigh,J.A. Sheikh, **N. T. Beigh**, Optimized MEMS Circular Membrane Resonators for Mechanical Energy Harvesting and Sensing for IoT Application. *In 2020 5th IEEE International Conference on Emerging Electronics (ICEE)*,(pp. **1-4**). IEEE(2020).[Link](#)
- [C22] F.T. Beigh; J. A. Sheikh; **N. T. Beigh**, A New Low Power Bi-Stable RF-MEMS Based Energy Harvester for NOMA Driven Industrial Internet of Things (IIoT). *In 2022 5th International Conference on Multimedia, Signal Processing and Communication Technologies (IMPACT)*,(pp. **1-8**).IEEE(2022). [Link](#)

REVIEWER ASSIGNMENTS

Nature Microsystems and Nanoengineering.
Nature Microsystem Technologies.
Nature Clean Technologies and Environmental Policy
Nature Journal of Inorganic and Organometallic Polymers and Materials.
IEEE Sensors Journal.
IEEE Sensor Conference 2023,2024,2025,2026.
ACS Applied Electronic Materials.
APL Electronic Devices.
MDPI Sensors.
Journal of Applied Physics.
Review of Scientific Instruments.
Journal of Climbing and Walking Robots

CERTIFICATIONS

Teaching certificate course-CTL Khalifa University
UAE Research Map Expert
Sustainability Foundations-Linkedin
IEEE Xtreme 14.0
NiPS Winter School-EnABLESEU
Multiphysics Modeling of Piezoelectric Sensors and Actuators-IEEE
3D Cable Modeling in COMSOL Multiphysics
Publishing Ethics Taylor & Francis Group
MATLAB-Mathworks
Elements of AI-UHR
The Hour of code

TALKS AND SEMINARS

July, 226 *Invited talk at Global Conference on Graphene and 2d Advanced Materials, Paris.*
Nov, 2025 *Invited talk at I2DM2025, UAE.*
Oct, 2025 *Invited talk at EUROSENSORS2025, Poland.*
Jan, 2025 *Invited talk at IWEHTECD2025, India.*
Oct, 2024, *Invited talk at NANO2024, UAE.*
Oct, 2022 *Invited talk at IEEE Sensors 2022 on High-Performance Piezoelectric Nanocomposite [Link](#)*
July, 2022 *Organized INUP i2i at NRF, IIT Delhi [Link](#)*
Dec, 2021 *Paper Presentation at IWPSD 2021 on SU-8/BTO Based Piezoelectric Nanogenerator*
Dec, 2021 *Paper Presentation at IEEE Transducers 2021 on Optimization Of ZnO/Su-8 nanocomposites.*
Dec, 2020 *Paper Presentation at IEEE-ICEE2020 on Parametrically Forced MEMS Energy Harvesters*
Dec, 2019 *Paper Presentation at IIT Indore on Design of Nonlinear MEMS Energy Harvesting.*
Dec, 2019 *Microsystems Technologies for Powering the 'Internet of Things-IoT' at Industry day, IIT Delhi*

HONOURS AND AWARDS

2026 UAE Research Map Expert.
2025 Gulf HE Scholarly Faculty.
2024 Best Thesis Award at IIT Delhi 2024 Convocation in IIT Delhi-2024.
2023 *Best Oral Presentation at SRS'23* in IIT Indore-2023.
2023 *Research Excellence Travel award (RETA)* in IIT Delhi-2023.
2023 *Best paper award* in 5th IEEE ICMSPECT 2023
2022 *International Travel Grant* by SERB-India for IEEE MEMS 2023
2022 *International Travel Grant* by SERB-India for IEEE Sensors 2022
2022 *International Travel Grant* in IEEE Sensors-2022.
2022 *Research Scholar Travel Award (RSTA)* in IIT Delhi-2022.
2022 *Gold Medal* in IIT Delhi Sportech-2022.
2019 *Best Poster Award in Clean & Renewable Energy* Industry Day 2019 at IIT Delhi.
2018 *3rd Rank* in the class of 60 students (M.Tech).
2018 *Gold Medal* in Inter University Soccer Tournament.
2017 *Scholarship* for scoring 95% in M.Tech.
2014 *Gold Medal* Inter University sketching Tournament

TECHNICAL SKILLS

Nano-Fabrication	CLEAN ROOM 100 <i>Mark SUSS MA6</i> and <i>ABM</i> Lithography Systems Direct Laser <i>MICROTECH DWL 66+</i> systems <i>ALCATEL</i> and <i>VECO</i> RF sputtering systems Spin coating systems(<i>SUSS+MicroTec</i>) RIE system(<i>HVV</i>) for silicon and oxide etching E-Beam Lithography (<i>Raith EBPG5200</i>) EHD Printing (Enjet-polymers) Voltera V-One DIW
Material Characterization	Piezoresponse Force Microscopy(<i>PFM</i>) with Cypher Kelvin Force Probe Microscopy(<i>KPFM</i>) with <i>AFM Bruker</i> Impedance measurements at LCR meter(<i>Hioki</i>) CV-IV-CF characterization of thin films at four probe systems
Device Characterization	Vibration Shaker acceleration tests Piezo-Tribo Imaprtter pressure tests Vector Network Analyzer (500kHz-18GHz)-Keithley Laser Doppler Vibrometer (Single Point-Polytech) Lock-in-Amplifier (SRS/Zurich) Universal testing machine (UTM-X350-20)
Modelling and Design	Multiphysics modelling in COMSOL and ANSYS Analytical modelling Mathamatica and MATLAB Photomask designing in L-Edit and Clewin
Circuit and PCB	Circut designing in MultiSim, LT-spice and Proteus PCB designing in KiCad and Eagle
Programming	Proficient in C, C++ Prior Experience in Python, VHDL, VERILOG
Others	Commercialization, Patents

LANGUAGE

- English, Hindi, Urdu :Fluent
- Arabic :Intermediate
- German :Pre-intermidiate
- Kashmiri : Mother tongue

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